

CURRICULUM VITAE
FOYEZ AHMMED
PHD CANDIDATE

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RESEARCH INTERESTS

My research focuses on both the development of statistical methods and the application of cutting-edge techniques to biomedical, healthcare, and public health investigations. Specifically, I am interested in the utilization of statistical methodology in cancer research, biomarkers discovery and evaluation, receiver operating characteristics (ROC), repeated measures and longitudinal analysis, causal inference, omics-based biomarker discovery including genomics, spatial transcriptomics, and proteomics, bioinformatics, machine learning, survival analysis, mixed-effects modeling, clinical trials, and Bayesian statistics, with demonstrated research experience through collaborations and publications.

EDUCATION

Doctor of Philosophy (Ph.D.) in Biostatistics <i>University of Kansas Medical Center, Kansas City, KS, USA</i>	Jan 2023 – Sep 23, 2026 (Defense)
Master of Science (M.S.) in Biostatistics <i>University of Kansas Medical Center, Kansas City, KS, USA</i>	Jan 2023 – May 2025 CGPA: 4.00/4.00
Master of Science (M.S.) in Statistics <i>Department of Statistics, University of Dhaka, Bangladesh</i>	Jan 2016 – Jul 2017 CGPA: 3.87/4.00
Bachelor of Science (B.S.) in Statistics <i>Department of Statistics, University of Dhaka, Bangladesh</i>	Jan 2011 – Jul 2015 CGPA: 3.71/4.00

ACADEMIC & RESEARCH POSITIONS

ORISE Research Fellow <i>U.S. Food and Drug Administration (FDA), USA</i>	Starting at Sep 28, 2026
Graduate Research Assistant <i>University of Kansas Medical Center, USA</i>	Jan 2023 – Present
Instructor (Biostatistics Tutoring Lab) <i>University of Kansas Medical Center, USA</i>	Spring 2025
Assistant Professor in Statistics <i>Comilla University, Bangladesh</i>	Jul 2022 – Dec 2022
Lecturer in Statistics <i>Comilla University, Bangladesh</i>	Jul 2019 – Jun 2022
Lecturer in Statistics <i>IUBAT, Bangladesh</i>	Jan 2019 – Apr 2019

RESEARCH WORKS

Published Journal Articles Google Scholar

- Hussain PM, Rahman MM, Kader MR, et al. (2026). Exploring the association between low birth weight and different forms of child malnutrition: A multi-country propensity score matching analysis.
Accepted in PLOS One.

2. Rayamajhi S, Sipes J, Ngala B, et al. (2026). Circulating extracellular vesicle protein biomarkers for the early detection of high-grade serous ovarian cancer. *Molecular & Cellular Proteomics*, 25, 101508.
3. Sipes J, Rayamajhi S, Bantis LE, et al. (2025). Spatial transcriptomic profiling of the human fallopian tube epithelium reveals region-specific gene expression patterns. *Communications Biology*, 8, 520.
4. Sipes J, Rayamajhi S, Bantis LE, et al. (2025). Defining the Ovarian Cancer Precancerous Landscape through Modeling Fallopian Tube Epithelium Reprogramming Driven by Extracellular Vesicles. *Cancer Research Communications*, 5(8), 1266–1281.
5. Kader MR, Rahman MM, Bristi PD, **Ahmed F** (2024). Change in mental health service utilization from pre- to post-COVID-19 period in the United States. *Heliyon*, 10(22), e40454.
6. **Ahmed F**, Hasan MN, Hossain MF, et al. (2024). Association between short birth spacing and child malnutrition in Bangladesh: a propensity score matching approach. *BMJ Paediatr Open*, 8(1), e002240.
7. **Ahmed F**, Hossain MJ, Khan MTF, et al. (2024). Mediating effect of BMI on the association of economic status and coexistence of hypertension and diabetes in Bangladesh: A counterfactual framework-based weighting approach. *Health Sci Rep*, 6, e2063.
8. Hossain MJ, Jabin N, **Ahmed F**, Sultana A, Abdur Rahman SM, Islam MR (2023). Irrational use of antibiotics and factors associated with antibiotic resistance: Findings from a cross-sectional study in Bangladesh. *Health Sci Rep.*, 6, e1465.
9. Al-Mamun M, Alam M, Hossain MJ, et al. (2023). Child drowning and associated risk factors: findings from a qualitative study in Bangladesh. *Health Science Reports*, 6, e1380.
10. Haque MA, Hossain MS, **Ahmed F**, et al. (2022). Abelmoschus esculentus (L.) Moench Pod Extract Revealed Antagonistic Effect against the Synergistic Antidiabetic Activity of Metformin and Acarbose upon Concomitant Administration in Glucose-Induced Hyperglycemic Mice. *Biologics*, 2(2), 128–138.
11. Hossain MJ, **Ahmed F**, Khan MR, Rashid PT, et al. (2022). Impact of Prolonged COVID-19 Lockdown on Body Mass Index, Eating Habits, and Physical Activity of University Students in Bangladesh: A Web-Based Cross-Sectional Study. *Front. Nutr*, 9, 873105.
12. Hossain MJ, **Ahmed F**, Khandokar L, Rahman SMA, Hridoy A, et al. (2022). Status of psychological health of students following the extended university closure in Bangladesh: Results from a web-based cross-sectional study. *PLOS Global Public Health*, 2(3), e0000315.
13. Haque AKM, Arifuzzaman BM, **Ahmed F**, et al. (2022). Semantic web in healthcare: a systematic literature review of application, research gap, and future research avenues. *International Journal of Clinical Practice*, 6807484.
14. **Ahmed F**, Hossain MJ, Sutopa TS, et al. (2022). The trend in exclusive breastfeeding practice and its association with maternal employment in Bangladesh: A multilevel analysis. *Front. Public Health*, 10, 988016.
15. **Ahmed F** (2022). Women's empowerment and practice of maternal healthcare facilities in Bangladesh: A trend analysis. *Journal of Health Research*, 36(6), 1104–1117.
16. Hossain MJ, **Ahmed F**, et al. (2022). Factors associated with underprivileged e-Learning, session jam phobia, and the subsequent mental distress among students following the extended university closure in Bangladesh. *Frontiers in Public Health*, 9, 807474.
17. Hossain MJ, Hridoy A, Rahman SA, **Ahmed F** (2021). Major depressive and generalized anxiety disorders among university students during the second wave of COVID-19 outbreak in Bangladesh. *Asia Pacific Journal of Public Health*, 10105395211014345.
18. Hossain MJ, **Ahmed F**, Rahman SA, Sanam S, Emran TB, Mitra S (2021). Impact of online education on fear of academic delay and psychological distress among university students following one year of COVID-19 outbreak in Bangladesh. *Heliyon*, 7(6), e07388.
19. Bari M, Hossain M, **Ahmed F**, et al. (2021). Knowledge, perception, and willingness towards immunization among Bangladeshi population during COVID-19 vaccine rolling period. *Vaccines*, 9(12), 1449.

20. **Ahmmmed F**, Manik MMR, Hossain MJ (2021). Caesarian section (CS) delivery in Bangladesh: A nationally representative cross-sectional study. *PLOS ONE*, 16(7), e0254777.
21. **Ahmmmed F**, Manik MMR (2021). Trends in early initiation of breastfeeding in Bangladesh and a multilevel analysis approach to find its determinants. *Scientific Reports*, 11(1), 1–9.
22. **Ahmmmed F**, Jamee AR (2021). Generalized quasi-likelihood estimation procedure for non-stationary over-dispersed longitudinal counts. *Journal of Statistical Computation and Simulation*, 91(9), 1802–1814.

Peer-Reviewed Conference Abstracts

1. Pham JM, Elsafy H, Alallaf J, et al. (2025). Exploring the significance of subclonal patterns of mismatch repair proteins in endometrial carcinomas. *Archives of Pathology & Laboratory Medicine*, 149(8). [Link](#)
2. Lee AA, Rayamajhi S, Hu M, et al. (2024). MicroRNA and proteomic analyses of plasma extracellular vesicles detect diffuse large B-cell lymphoma. *Clinical Cancer Research*, 30(21). [Link](#)

Manuscripts Under Review

1. **Ahmmmed F**, Bantis LE. Quantifying the complementary value of continuous correlated biomarkers using the overlapping volume of their bivariate densities: A generalized ROC framework on $\mathbb{R}^2$.
Under review in the Journal of Biopharmaceutical Statistics

Manuscripts Under Preparation

1. **Ahmmmed F**, Bantis LE. A generalized ROC framework on $\mathbb{R}^2$ for combining biomarkers using a cost-sensitive Bayes decision rule.
Target journal: *Biometrics*.
2. Lee AA, Rayamajhi S, Hu M, et al. MicroRNA and proteomic analyses of plasma extracellular vesicles detect diffuse large B-cell lymphoma.
Target journal: *Communications Biology*.

R and MATLAB Packages

As part of my dissertation research, I have developed R and MATLAB packages. These packages will be submitted for public release upon acceptance of the associated manuscript.

COURSES TAUGHT

- **Department of Statistics, Comilla University, Bangladesh**
 - *Graduate Program (MSc)*
 - Advanced Statistical Inference
 - Advanced Survival Analysis
 - *Undergraduate Program (BSc)*
 - Sampling Distribution
 - Calculus-I
 - Stochastic Process
 - Mathematical Demography
- **Department of Mathematics/Physics, Comilla University, Bangladesh**
 - *Undergraduate Program (BSc)*
 - Basic Statistics (1st year)

CONFERENCE PRESENTATIONS / WORKSHOPS

1. **Ahmmmed F**, Bantis LE. Quantifying the complementary value of continuous correlated biomarkers using the overlapping volume of their bivariate densities: A generalized ROC framework on $\mathbb{R}^2$.
 - *18th Annual IDAD: Frontiers in Biostatistics & Data Science Conference, 2025.*

- *University of Kansas Cancer Center Research Symposium, 2025.*
- *American Statistical Association, Kansas–Western Missouri Chapter Meeting, 2024.*
- *17th Annual IDAD: Frontiers in Biostatistics & Data Science Conference, 2024.*
- *University of Kansas Cancer Center Research Symposium, 2024.*

2. Ingenuity Pathway Analysis (IPA) Training Workshop

QIAGEN Digital Insights, Lied Auditorium, University of Kansas Medical Center, January 19, 2024.

3. Scientific Writing Workshop: “Reader Expectations Approach to Scientific Writing”

University of Kansas Medical Center, KU Edwards Campus (Regnier Auditorium), January 25–26, 2024.

SKILLS

Programming Language:	R, MATLAB, Python & SQL
Statistical Software:	SAS, SPSS, Stata & OpenBUGS
Scientific Writing & Reporting:	R Markdown, MATLAB Live Scripts & LaTeX
Cloud Computing:	High Performance Clustering (HPC)
Version Control:	Git, GitHub

AWARDS, FELLOWSHIP & RESEARCH GRANTS

Vice Chancellor Research Award.

Comilla University, Cumilla-3506, Bangladesh 2023

National Science & Technology (NST) Fellowship, Bangladesh 2016

Merit Scholarship: Amar Ekushey Hall Trust Fund Scholarship based on M.S. result, University of Dhaka, Bangladesh 2016

Merit Scholarship: Amar Ekushey Hall Trust Fund Scholarship based on B.S. results, University of Dhaka, Bangladesh 2013–2015

RESEARCH GRANTS

BDT 110000, *University Grants Commission & Comilla University, Cumilla, Bangladesh* 2021–2022

BDT 95000, *University Grants Commission & Comilla University, Cumilla, Bangladesh* 2020–2021

BDT 110000, *University Grants Commission & Comilla University, Cumilla, Bangladesh* 2019–2020

SOME RESEARCH STUDENTS SUPERVISED

1. **Asaduzzaman (Session: 2015–2016).** “Effect of lockdown on the mental health of university students in Bangladesh due to COVID-19.” Department of Statistics, Comilla University, Bangladesh, 2021.
2. **Riaz Razu (Session: 2015–2016).** “Antenatal care visits in Bangladesh and its associated factors: A nationally representative cross-sectional study.” Department of Statistics, Comilla University, Bangladesh, 2021.
3. **Mohammad Fazle Rabbi (Session: 2019–2020).** “Factors associated with postnatal Care and facility delivery in Bangladesh based on BDHS data 2017–18.” Department of Statistics, Comilla University, Bangladesh, 2022.
4. **Jerin Tasnim (Session: 2019–2020).** “Association of overweight and obesity with Hypertension and Diabetes among adult in Bangladesh.” Department of Statistics, Comilla University, Bangladesh, 2022.

5. **Khadiza Akter (Session: 2019–2020).** “Factor Associated with Acute Respiratory Tract Infection among children in Bangladesh.” Department of Statistics, Comilla University, Bangladesh, 2022.

PROFESSIONAL TRAINING

- **Outcome-Based Education (OBE) Curriculum Design Training**
 - Institutional Quality Assurance Cell (IQAC), Comilla University, Bangladesh **2021**
 - Institutional Quality Assurance Cell (IQAC), International University of Business Agriculture and Technology, Bangladesh **2019**

PROFESSIONAL SERVICES, ACTIVITIES & MEMBERSHIP

- Student advisor and organizing committee member of the “Statistics Week,” Department of Statistics, Comilla University, Bangladesh. **2020–2022**
- Member of the syllabus design committee in both undergraduate and graduate programs at the Department of Statistics, Comilla University, Bangladesh. **2020–2022**
- Member of different examination committees of the Department of Statistics, Comilla University, Bangladesh. **2020–2022**
- Manager of soccer and cricket teams, Department of Statistics, Comilla University, Bangladesh. **2020–2022**
- Member of American Statistical Association (ASA). **2024–Present**
- Involved in different committees for successfully organizing seminars and conferences on Statistics, Comilla University, Bangladesh.
- **Reviews of Journal Articles:** *Journal of Medical Virology, Heliyon, BMC Digital Health, Journal of Health, Population and Nutrition, BMC Public Health, and Scientific Report.*